

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: SundayWasters

Project / Product Name: Agentic Code Retrieval (Theme 01: Agentic Code Intelligence)

Organization / Institution (if any): VIT Vellore

Submission Date: 4 October 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? **Yes**

We used AI coding agents throughout development. Claude Code (Anthropic; main session on Claude Sonnet 5.5, with Claude Opus subagents) wrote most of the code, experiment scripts, tests and documentation from our instructions, and three reviewer agents (the “council”: Sol and Astra = OpenAI Codex (GPT), Opus = Claude Opus) reviewed our plans and experiment results. The team set the direction, chose between options, decided every pipeline, read the results and tested the demo by hand. The retrieval system itself runs only local open-weight embedding models (EmbeddingGemma-300m and jina-code-embeddings-0.5b); no generative AI and no hosted API is used at query time.

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: Yes. The reviewer agents critiqued plans and proposed options (an exact-literal keyword lane, test-file scope, merging tiny fragments) and advised against a LightGBM or cross-encoder reranker. Claude Code proposed the learned score fusion. The two-stage idea (Gemma retrieves, jina-code chooses) and the move to EmbeddingGemma came from the team.

Code generation or assistance: Yes. Claude Code wrote most source code (tree-sitter chunker, vector cache and FAISS indexes, BM25 lane, fusion search, MTEB adapter, experiment scripts, Reflex demo) from our instructions; we reviewed it, ran it and asked for changes.

UI / UX design: Yes. Claude Code built the Reflex demo UI with its Reflex and design-taste-frontend skills; the layout was revised over several rounds of our feedback (page width, padding, title, results dropdown, removed sections).

Content creation: Yes. The README, architecture and handoff docs, the experiment log (docs/experiments/) and the two architecture diagrams (diagram-maker skill) were drafted by Claude Code and revised on our instructions.

Data analysis: Yes. Claude Code ran the benchmark and repository experiments and produced the comparison tables and per-category breakdowns; the reviewer agents judged what 50 questions can resolve. We read the results and made the decisions.

Testing / debugging: Yes. Claude Code wrote the pytest suite (no models or GPU needed) and fixed problems found during runs (a tree-sitter 0.26 crash, .d.ts files crowding results, an empty-cache bug, GPU out-of-memory errors, NaN output from one OpenVINO wrapper). We tested the demo by hand.

Other: Evaluation data. The labelled question sets for Node-RED and BrowserOS (50 questions each) were written by Claude Code / a Claude subagent from the source code, before any retrieval results were seen.

4. Feature Origin Classification (Please add/delete based on the number of the feature)

Feature 1

1. Feature Name: **Tree-sitter code chunker and file discovery**

2. Self-Generated/AI-Generated/Both: Both

3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification):

AI tools / platform: Claude Code (Anthropic; main session on Claude Sonnet 5.5, with Claude Opus subagents).

Prompt used: Given in an earlier Claude Code session (transcript not retained). Later change: “run the label check and the lexical lane and merge tiny fragments, then re-index.”

Output summary: Splits Python, JavaScript, TypeScript and TSX into function, method, class-header and statement-group units (at most 2,500 characters) with exact file:line spans.

Modification: On our direction: TypeScript/TSX added, .d.ts and generated/ files skipped, fragments under 60 characters merged, tree-sitter pinned below 0.26 after it crashed on real JavaScript.

Feature 2

1. Feature Name: **Embedding model choice and official benchmark path (MTEB AppsRetrieval)**

2. Self-Generated/AI-Generated/Both: Both

3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification):

AI tools / platform: Claude Code (Anthropic; main session on Claude Sonnet 5.5, with Claude Opus subagents); reviewer agents (the “council”): Sol and Astra = OpenAI Codex (GPT), Opus = Claude Opus.

Prompt used: “how do these numbers that we've got compare to base models on the same test set?”; “yes, run embeddinggemma, jina-code and qwen3-0.6b on the

test split”; “i want to change from jina-code to embeddinggemma-300m ... discuss that with the council as well”.

Output summary: PrePostPipelineEncoder adapter and run script; official-format runs: EmbeddingGemma 84.28, jina-code 83.83, Qwen3-Embedding-0.6B 73.49 NDCG@10.

Modification: The official path stays a plain encoder; the team dropped Qwen3 and had the optional OpenVINO int8 path removed (“let's not use int8 export models”).

Feature 3

1. Feature Name: **Gated BM25 keyword lane and test-file scope**

2. Self-Generated/AI-Generated/Both: Both

3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification):

AI tools / platform: Claude Code (Anthropic; main session on Claude Sonnet 5.5, with Claude Opus subagents); reviewer agents (two Codex agents and Claude Opus) identified weak exact-literal ranking and suggested a gated lane.

Prompt used: “commit current changes; then run the label check and the lexical lane and merge tiny fragments, then re-index.”

Output summary: Identifier-aware BM25 that runs only when a question names an identifier or quotes a message found in the code, fused by reciprocal rank fusion; test files excluded unless asked for. Node-RED MRR@10 0.755 to 0.848, BrowserOS 0.707 to 0.734.

Modification: The team chose which variants to test and kept the lane for repositories only (it lowers the AppsRetrieval score).

Feature 4

1. Feature Name: **Two-stage retrieval (Gemma retrieves, jina-code re-scores)**

2. Self-Generated/AI-Generated/Both: Both (idea from the team)

3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification):

AI tools / platform: Claude Code (Anthropic; main session on Claude Sonnet 5.5, with Claude Opus subagents).

Prompt used: “since gemma is the better chooser, but jina is the better selector, can we sort of give the top-10 selections of gemma to jina to choose one?”; “run this two stage pipeline on cpu alone.”

Output summary: Two-stage search and CPU latency benchmark; it helped on BrowserOS and hurt on Node-RED, with an analysis of why.

Modification: The team asked to widen the pool from 10 to 50 and combine it with score fusion (next feature).

Feature 5

1. Feature Name: **Learned score fusion (final ranking)**

2. Self-Generated/AI-Generated/Both: Both

3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification):

AI tools / platform: Claude Code (proposed the idea); the Claude Opus reviewer checked feasibility.

Prompt used: “i like your idea about the learned score fusion, ask the opus council member ... specifically if that's feasible”; “try the learned fusion score approach on two stage approach? ... should we try and change the top-10 from gemma to top-50?”

Output summary: Gemma's top 50 re-scored by $(1 - g)(0.7 \text{ jina} + 0.3 \text{ Gemma}) + g \text{ BM25}$, with the weights chosen on one repository and tested on the other. Node-RED MRR@10 0.908, BrowserOS 0.777.

Modification: The team rejected a LightGBM or cross-encoder reranker after the reviewers' advice and settled on this pipeline (“the learned fusion approach seems to work. So let's leave it at that.”).

Feature 6

1. Feature Name: **Labelled evaluation question sets (Node-RED, BrowserOS)**

2. Self-Generated/AI-Generated/Both: AI-Generated (team specification)

3. Description (include AI tools/platform Used, prompt used, Output Summary, Modification):

AI tools / platform: Claude Code and a Claude subagent.

Prompt used: “clone this repository under eval/ ... write about 50 queries on this repository, with exact and behavioural testing queries in the form of actual questions (not just sentences), with the same split ... use a subagent for the creation of these queries”.

Output summary: 50 questions per repository (30 dev, 20 holdout) with gold file:line ranges, written before any retrieval results were seen.

Modification: Two Node-RED labels were corrected after an audit; the change is recorded in the file (label_fixes).

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Deep Arjit Prasad

Role: Team Member

Signature: 

